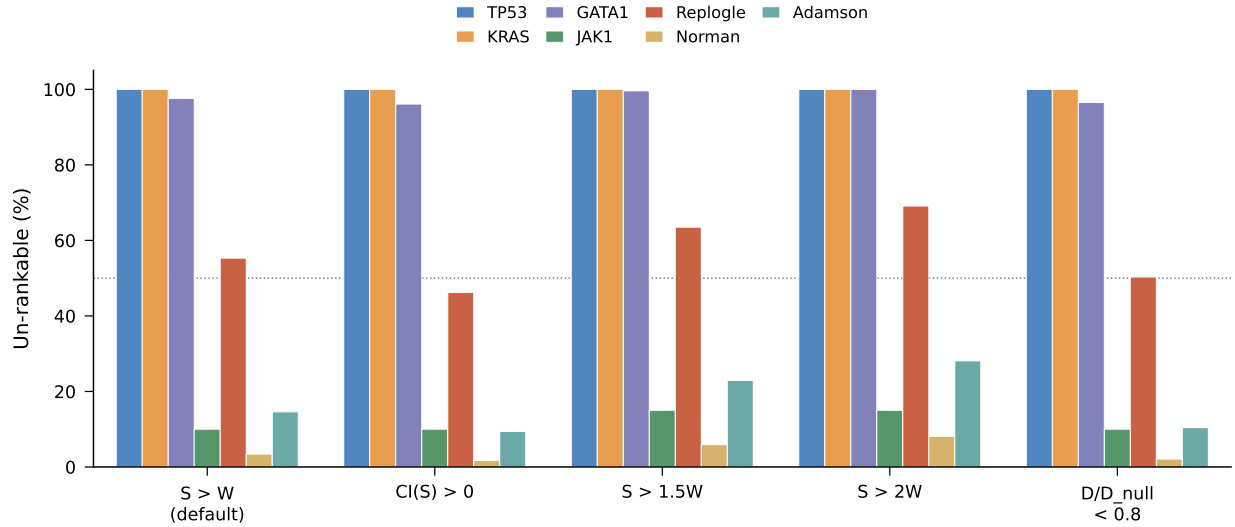


Supplementary Information

AllelePerturb reveals measurement limits of protein-coding variant prediction in single-cell transcriptomics

This document contains the full parameter tables underlying the variant feature representation, the prediction methods and the split-half rankability analysis. Complete assignments and code are also available at <https://github.com/Boom5426/AllelePerturb>.



Supplementary Figure 1: **The rankability verdict is robust to the choice of criterion.** Fraction of un-rankable perturbations for the seven datasets (the four AllelePerturb genes and three gene-level Perturb-seq atlases) under five rankability criteria: the primary signal-versus-noise rule $S > W$ (default); a bootstrap lower-bound rule ($CI(S) > 0$, the lower 95% confidence bound of S above zero); two increased-stringency rules ($S > 1.5W$ and $S > 2W$); and a ratio threshold, under which a variant is un-rankable when $D_{self}/D_{null} \geq 0.8$. Absolute fractions shift with stringency, but the ordering across datasets is preserved under every criterion: TP53 and KRAS remain $\sim 100\%$ un-rankable, GATA1 stays high, JAK1 stays low ($\sim 10\%$), and among gene-level atlases Replogle is consistently the highest and Norman the lowest. Source values are listed in `results/rankability_sensitivity.csv`.

Supplementary Table 1: **Variant feature (θ) structural and hotspot annotations.** Per-gene residue assignments for the three structural components of the biophysical feature vector. The fold-core term f_v was additionally set to 0.5 for annotated interaction regions and 0.0 elsewhere. The hotspot/pathogenic marker h_v was defined exclusively from external prior knowledge, with no reference to the single-cell outcome.

Gene	Structural core ($f_v = 1$)	Catalytic switch ($s_v = 1$)	/	Hotspot / pathogenic ($h_v = 1$)
TP53	DNA-binding domain, residues 94–293	—		COSMIC/IARC missense hotspot codons 175, 176, 179, 220, 238, 245, 248, 249, 273, 282, 285
KRAS	G-domain, residues 1–169	G12, G13, Q61		COSMIC/IARC missense hotspot codons 12, 13, 59, 61, 117, 146
GATA1	Zinc fingers ZF1 (204–228) and ZF2 (258–282)	Zinc-finger cysteines		Zinc-coordinating cysteines 204, 207, 225, 228, 258, 261, 279, 282; ClinVar pathogenic / likely-pathogenic annotations
JAK1	Kinase domain, residues 866–1154	—		Functional-module membership: JH2 pseudokinase (583–855), JH1 kinase (875–1153)

Note. An earlier version of h_v used the top quintile of measured effect size, which leaks the evaluation outcome into the feature; it was removed and the entire evaluation grid was re-run, with ranking performance unchanged (θ -only PDS shifted by at most 0.02 and every method remained at chance).

Supplementary Table 2: **Regression-head hyperparameters.** The six heads were crossed with three feature spaces (θ , ESM, ESM+ θ) to form the 18 in-house feature-model combinations. All heads used a fixed random seed and were fit by minimizing squared error on the training variants.

Head	Family	Settings
Ridge regression	Linear (L2)	$\alpha = 1.0$
Lasso regression	Linear (L1)	$\alpha = 0.01$; up to 2,000 iterations
Random forest	Tree ensemble	50 trees; maximum depth 6
Gradient-boosted trees	Tree ensemble	50 estimators; maximum depth 3; multi-output wrapper
k -nearest-neighbours	Neighbour	$k = 3$
Multilayer perceptron	Neural network	hidden layers 128 and 64; ReLU; up to 500 iterations

Note. Linear and neighbour heads regress directly onto the full expression profile. Because tree ensembles scale poorly to thousands of output genes, for output spaces larger than 200 genes the random forest and gradient-boosted heads were fit on a q -component PCA of the training

profiles and predictions were mapped back to gene space by the inverse transform, with $q = \min(50, G, n_{\text{train}} - 1)$.

Supplementary Table 3: **Split-half rankability and power-analysis sampling grid.** Parameters for the energy-distance split-half analysis (Fig. 3) and the depth power analysis (Figs. 3, 5). Each configuration was summarized by the median and the 2.5–97.5 percentile interval across seeds.

Parameter	Values
Split-half subsample sizes (per half)	30, 50, 100, 150, 250, 365 cells
Representation spaces	50-component PCA (canonical), gene space, pathway-score space
Random seeds per configuration	50
Canonical setting	PCA space at native maximum depth
Power-analysis depths (per variant)	50, 100, 150, 300 cells
Minimum cells for inclusion	50 (split-half); 5 (pseudobulk profiles)

Supplementary Table 4: **Metric-family invariance of the measurement floor.** Split-half window ratio (median D_{self} divided by median D_{null}) per gene under five distance families in the 50-component PCA space, at 50 cells per half. A ratio near 1 indicates a variant at the measurement floor; a low ratio indicates a wide, resolvable window. The floor is present for TP53, KRAS and GATA1 and absent for JAK1 under every metric.

Gene	Energy	MMD (RBF)	Sliced-Wasserstein	Cosine	Euclidean
TP53	1.03	1.01	1.02	1.02	1.03
KRAS	0.95	0.97	0.97	0.99	0.97
GATA1	1.01	0.98	1.03	0.90	1.04
JAK1	0.12	0.32	0.40	0.05	0.26

Supplementary Table 5: **Benchmark resolution across public perturbation datasets and the allele-level genes.** Resolution is the probability that a benchmark recovers the true ranking of graded synthetic predictors under bootstrap over perturbations (the controlled-predictor analysis of the main text), computed at 50 cells per half over perturbations with at least 100 cells; allele perturbation counts and rankable fractions therefore differ from the native-depth main-text table. Well-powered gene- and drug-level benchmarks recover model orderings, whereas TP53 and KRAS are the only datasets at the floor; resolution reflects both per-perturbation signal and the number of perturbations.

Dataset	Type	Perturbations	Rankable	Resolution
Norman 2019	gene pairs	224	74%	1.00
Repogle 2022	gene KD	400	33%	1.00
Adamson 2016	gene KD	97	42%	1.00
Virtual Cell Challenge	gene KD	145	22%	1.00
sci-Plex	drug	188	2%	0.77
JAK1	allele	14	86%	0.99
GATA1	allele	247	0%	0.51
KRAS	allele	92	0%	0.02
TP53	allele	98	1%	0.01

Supplementary Table 6: **Gene-shared versus allele-specific decomposition.** Each variant effect is split as $\delta_v = \bar{\delta}_g + r_v$. Columns give the fraction of the per-variant effect that is gene-shared versus the allele-specific residual, and the oracle discrimination ceiling scored on the full profiles versus on the residual (allele-specific) profiles. Across all 24 predictors (the 18 feature-model combinations, the Gene-mean reference and the five published variant-conditionable models) the residual Pearson- δ lay between -0.08 and 0.03 , so direction recovery is essentially entirely gene-shared. The oracle columns are recomputed under the residual-decomposition protocol and differ slightly from the per-split canonical ceiling reported in the main text.

Gene	Gene-shared	Allele residual	Oracle (full)	Oracle (residual)
TP53	72%	28%	0.488	0.480
KRAS	59%	41%	0.481	0.478
GATA1	29%	71%	0.573	0.588
JAK1	51%	49%	0.808	0.892

Supplementary Table 7: **Representation robustness control.** PDS-cosine and Pearson- δ for each variant representation under the identical splits, six regression heads and scoring (single deterministic pseudobulk draw; the multi-seed convention used in the main text gives PDS 0.49–0.52). Modern sequence embeddings, mutation-aware mutant-minus-wild-type constructions, and a structure- and evolution-informed variant-effect predictor (AlphaMissense) all leave allele discrimination at chance while recovering direction similarly. A featureless training-gene-mean predictor reaches Pearson- δ 0.66 at PDS 0.45.

Representation	PDS-cosine	Pearson- δ
θ (biophysical, 6-d)	0.475	0.632
ESM-1v (1,280-d)	0.464	0.640
ESM2 global	0.465	0.645
ESM2 mutant–WT global delta	0.465	0.646
ESM2 mutated-residue delta	0.468	0.631
ESM2 local-window delta	0.466	0.646
ESM2 residue delta + θ	0.465	0.627
AlphaMissense	0.459	0.655
AlphaMissense + θ	0.466	0.656

Supplementary Table 8: **Oracle ceiling sensitivity.** The replicate-based discrimination ceiling re-computed over all variants of each gene under stress tests (small differences from the main-text per-split values reflect the all-variant candidate set). Full depth uses all cells per half; perfect prediction uses a deep disjoint estimate of the true mean scored against the benchmark’s noisy pseudobulk truth; shrinkage reports a prediction shrunk fully to the gene mean ($\lambda=0$) versus the replicate oracle ($\lambda=1$). The ceiling is unchanged by depth and stays at chance for TP53 and KRAS even for a perfect prediction; shrinking toward the gene mean does not raise it.

Gene	Oracle (300-cell)	Oracle (full depth)	Perfect vs noisy	Shrink to gene-mean	Replicate oracle
TP53	0.487	0.486	0.488	0.500	0.487
KRAS	0.485	0.492	0.490	0.500	0.485
GATA1	0.585	0.591	0.569	0.500	0.585
JAK1	0.812	0.812	0.997*	0.500	0.812

*JAK1 perfect-versus-noisy is computed over only the few JAK1 variants with more than 350 cells and is not representative of the gene.

Supplementary Table 9: **A decision table for allele-resolution perturbation prediction.** Three nested questions of increasing difficulty locate a dataset by its failure mode: detection (a variant versus wild type; assay sensitivity), identification (one variant versus another; ground-truth resolution) and prediction (a held-out variant from its features; computational generalization). Prediction is only meaningful once identification is achievable.

Detection	Identification	Prediction	Limiting regime	In AllelePerturb
no	—	—	assay-limited	(none)
yes	no	not evaluable	resolution-limited	TP53, KRAS
yes	partial	not evaluable	resolution-limited	GATA1
yes	yes	no	model-limited	JAK1
yes	yes	yes	successful regime	(target)

Supplementary Table 10: **Published perturbation models under a unified allele-level evaluation.** Perturbation-discrimination score (PDS, cosine) with 95% bootstrap confidence intervals, pooled over five generalization splits and four genes; chance is 0.5. Variant-conditionable models are scored directly; gene-keyed models are allele-blind by construction and reported by resolution coverage. Every method was scored through one harness (identical held-out variants, pseudobulk profiles, tie-aware PDS and permutation null).

Model family	Method	PDS (95% CI)	Pearson- δ
<i>Variant-conditionable (run to completion)</i>			
Optimal-transport flow	CellFlow	0.51 (0.48–0.54)	0.12
Attribute disentangle- ment	Biolord	0.49 (0.47–0.52)	0.13
Latent arithmetic (dose)	scVIDR	0.50 (0.48–0.53)	0.13
Latent arithmetic	scGen	0.49 (0.46–0.52)	0.18
Conditional-flow gener- ative	PerturbNet [†]	0.54 (0.51–0.57)	0.13
Feature regression	baselines (Fig. 2)	0.49–0.52	0.55–0.65
<i>References</i>			
Gene identity	Gene-mean	0.49 (0.46–0.52)	0.66
No perturbation	WT-null	0.50	—
<i>Allele-blind by construction (resolution coverage 4/470 = 0.85%)</i>			
Foundation model	scGPT	n/a [‡]	—
Gene knowledge graph	GEARS	n/a [‡]	—
Virtual-cell (one-hot)	STATE	n/a [‡]	—
<i>Did not converge</i>			
Compositional autoen- coder	variant-CPA	non-finite loss	—

[†]PerturbNet generates within a 50-dimensional principal-component subspace in which random predictions already score 0.52; full-space-at-chance predictors (Ridge-esm, Gene-mean) projected into the same subspace are likewise inflated to 0.52, and its exceedance over a subspace permutation null is not significant (permutation $P = 0.13$). [‡]These models condition on perturbed-gene identity and assign an identical prediction to every variant of a gene, so an allele-level PDS is undefined (a category error).

Supplementary Table 11: **Per-gene measurement window, achievable ceiling and model performance.** Split-half energy-distance ratio $D_{\text{self}}/D_{\text{null}}$; oracle discrimination ceiling (independent second measurement, estimating attainable discrimination); fraction of allele pairs resolvable above replicate noise; fraction of variants rankable at native depth; and best honestly-scored model PDS (canonical multi-seed harness). Chance is 0.5.

Gene	$D_{\text{self}}/D_{\text{null}}$	Oracle ceiling (95% CI)	Pairs resolvable	Rankable (native)	Best model PDS [†]
TP53	0.96	0.485 (0.445–0.525)	0%	0%	0.51
KRAS	1.00	0.500 (0.440–0.566)	0%	0%	0.54
GATA1	0.88	0.572 (0.549–0.595)	3.8%	2.4%	0.50
JAK1	0.21	0.792 (0.742–0.838)	78.5%	90%	0.52

[†]Best model is the highest-scoring in-house feature-regression head per gene under the canonical multi-seed harness (published variant-conditionable models are compared separately in Supplementary Table 10). PerturbNet is excluded; its higher per-gene values (0.53–0.54) arise from its 50-dimensional principal-component subspace, in which random predictions already score 0.52.